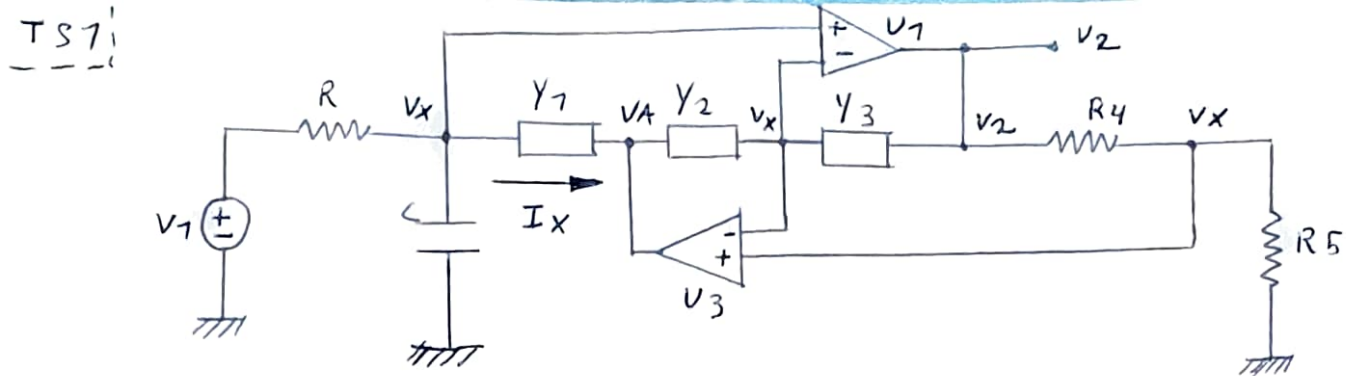




$$\rightarrow (v_2 - v_x) G_4 = v_x G_5 \rightarrow v_2 \cdot G_4 = v_x (G_4 + G_5) \rightarrow \boxed{v_2 = v_x \left(1 + \frac{G_5}{G_4} \right)} \quad (1)$$

$$\boxed{(v_A - v_x) Y_2 = (v_x - v_2) Y_3} \quad (2)$$

$$(1) \rightarrow (2): (v_A - v_x) Y_2 = \left(\cancel{v_x} - \cancel{v_x} - v_x \frac{G_5}{G_4} \right)$$

$$v_A \cdot Y_2 - v_x \cdot Y_2 = -v_x \frac{Y_3 \cdot G_5}{G_4}$$

$$v_A \cdot Y_2 = v_x \left(Y_2 - \frac{Y_3 \cdot G_5}{G_4} \right) \rightarrow \boxed{v_A = v_x \left(1 - \frac{Y_3 \cdot G_5}{Y_2 G_4} \right)} \quad (3)$$

$$\boxed{I_x = (v_x - v_A) Y_1} \quad (4)$$

$$\boxed{Y_e = \frac{I_x}{v_x}} \quad (5)$$

$$(3) \rightarrow (4): I_x = \left(\cancel{v_x} - \cancel{v_x} + v_x \cdot \frac{Y_3 G_5}{Y_2 G_4} \right) Y_1 \rightarrow \boxed{I_x = v_x \cdot \frac{Y_1 Y_3 G_5}{Y_2 G_4}} \quad (6)$$

$$(6) \rightarrow (5): Y_e = \frac{\cancel{v_x}}{\cancel{v_x}} \cdot \frac{Y_1 Y_3 G_5}{Y_2 G_4} \rightarrow \boxed{Y_e = \frac{Y_1 Y_3 G_5}{Y_2 G_4}}$$

$$\rightarrow \boxed{Z_e = \frac{R_5 Z_7 Z_3}{R_4 Z_2}}$$